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Geriatrics

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SUMMARY PEARLS

- By 2030, one in six people in the world will be age 60 years or older. Physiatrists can play a large role in maintaining function and independence of older individuals.
- Sarcopenia is a progressive and generalized skeletal muscle disorder associated with increased likelihood of adverse outcomes. The diagnosis of sarcopenia is based on muscle strength (hand grip <27 kg for males, <16 kg for females), muscle quality, and physical performance (e.g., gait speed <0.8 m/sec).
- Falls in the elderly can have a single or multifactorial cause. Prevention is the best strategy, but it is important to thoroughly investigate possible contributing factors and address them after a fall.
- The Institute for Healthcare Improvement developed the Age-Friendly Health Systems Initiative to guide geriatric care using the 4Ms: Medication, Mentation, Mobility, What Matters to You.
- Medication reconciliation should take place during every encounter as an opportunity to deprescribe, downtitrate, or further educate a patient about the medication's proper usage.
- Delirium can be prevented or addressed early through improved recognition by all members of the care team.
- Mobility goals should be determined using realistic expectations for each individual patient according to their own goals for activity and quality of life.
- Providers who regularly take care of geriatric patients will often be challenged with biomedical ethics conflicts. These providers should work to understand the basic principles of biomedical ethics and how to utilize an ethical decision-making model.

Epidemiology and Physiologic Changes of Aging

People are living longer worldwide. Consequently, countries are experiencing population growth in total number as well as the proportion of older adults. It is projected that by 2030, one in six people in the world will be aged 60 years or over.⁵⁷ Between 2015 and 2050, the proportion of the world's population over 60 years old will nearly double from 12% to 22%. There are numerous health conditions and physiologic changes associated with aging. As people age, their body undergoes changes in different areas often at the same time.⁵⁷ The details of the physiologic changes associated with aging are summarized in Table 31.1. Mobility decline with aging is an essential area to address in maintaining a person's independence and is one area in which physiatrists are often consulted. Sarcopenia (reduction in muscle mass, strength, and power) is one of the main culprits of mobility decline, which is elaborated on in the next section.

Sarcopenia

Sarcopenia is a progressive and generalized skeletal muscle disorder that is associated with an increased likelihood of adverse outcomes, including falls, fractures, physical disability, and mortality.⁴³ Sarcopenia has been an exciting area of basic and clinical research due to its predictability of adverse outcomes, its responsiveness to treatment, and the associated cost. Sarcopenia is

formally listed as muscle disease with a 2024 International Classification of Diseases, Tenth Revision, Clinical Modification code M62.84 (icd10data.com). However, there has been a lag in the implementation of recognition and treatment of sarcopenia in clinical practice. The focus of practice should be on preventing, delaying, treating, and even reversing sarcopenia by systematic screening and interventions with the aims of improving mobility and decreasing falls, hospitalizations, and deaths.

Several organizations and scientific societies have developed their own sarcopenia definitions specific to their regions. To develop an international consensus on the definition, the Global Leadership Initiative in Sarcopenia (GLIS) was formed in 2019 to 2021.²⁶ The GLIS has developed a conceptual definition of sarcopenia, which will be used to formulate an operational definition of the term. GLIS agreed that muscle strength, muscle-specific strength (muscle strength normalized to the cross-sectional area of a muscle or muscle group), and muscle mass are components of sarcopenia, and physical performance (e.g., gait speed) is the outcome rather than a component of sarcopenia. Until a new operational definition by GLIS is available, the definition from the European Working Group on Sarcopenia in Older People 2 (EWGSOP2) revised in 2018 and published in 2019 is applicable.¹³ The revised definition from EWGSOP2 focuses on muscle function rather than muscle mass based on the better predictability of muscle function on adverse outcomes compared to muscle mass. The EWGSOP2 cutoff points are as follows: (1) muscle strength: five times chair rise >15 sec or hand grip strength

TABLE 31.1 Age-Related Changes in Multiple Organs and Systems

Organ/System	Age-Related Changes
Musculoskeletal system	1.5%–2% decline in muscle strength/year after age 40 years Preferential loss of type II fibers Decline in muscle power more pronounced than muscle strength Bone changes: bone mineral density loss, postural changes (see more details in the text)
Neurologic system	Depletion of neurons in brain, loss of anterior horn cells Decreased nerve conduction velocities
Cardiovascular system	Cardiac output decreases after third decade at a rate of 1%/year Causes for ↓ in cardiac output: ↓ in inotropic response to catecholamines, ↑ in myocardial stiffness, and progressive stiffening of the aorta (↑ afterload) Decrease in aerobic capacity (↓VO ₂ max)
Respiratory system	Linear ↓ in vital capacity: males (26 mL/year), females (22 mL/year) after age 20 years; residual volume (RV) ↑ with age Total lung capacity (TLC) remains constant RV/TLC ↑ from 20% at age 20 years to 35% at age 60 years ↑ in alveolar-arterial oxygen difference O ₂ due to mismatch of ventilation and perfusion <i>Predisposition to pneumonia due to depression of immune system function, aspiration of oropharyngeal secretions, slower mechanical clearing of tracheobronchial tree by mucociliary apparatus, less effective cough</i>
Kidney/urinary system	↓ in number of glomeruli (1,000,000 at age 30 years to 700,000 by age 65 years) ↓ in creatinine clearance ↓ in glomerular filtration rate: 10%/10 years after age 30 years No change in serum creatinine concentration with age due to age-related ↓ in creatinine production (↓ in muscle mass) <i>Drugs cleared by glomerular filtration (e.g., aminoglycosides, digoxin, penicillin, tetracycline) may have prolonged half-life even if the dosage is adjusted based on serum creatinine concentration.</i> ↓ in ability to concentrate and dilute urine: ↑ vulnerability for dehydration and hyponatremia. Urinary incontinence (17% of males, 23% of females age >65 years), and overflow incontinence due to possibly reduced and delayed sensation of urge to urinate
Gastrointestinal system	Presbyesophagus: ↓ in peristaltic response, ↑ in nonperistaltic response, a delayed transit time ↓ in relaxation of the lower sphincter during swallowing; may lead to dysphagia and reduced calorie consumption Atrophic gastritis (40% of those age ≥65 years): risk of achlorhydria, deficient intrinsic factor secretion, ↓ pepsinogen production Hypotonic colon: longer stool transit time, dehydration, which eventually causes chronic constipation Loss of control in internal and external anal sphincters Prevalent diverticulosis (50% in those age ≥80 years) Liver function shows little change with aging, yet drugs (e.g., diazepam) are metabolized more slowly with aging
Endocrine system	↓ in number and function of insulin-producing beta cells More inactive precursor proinsulin released in response to a glucose challenge and development of progressive peripheral insulin resistance with age ↓ in lean body mass with relative ↑ in adiposity (↑ in fat cell size, not in number of fat cells)
Sexual function	Males produce less sperm, sexual responses are delayed by age 50 years. Impotence in males is <i>not</i> normal part of aging. Females after menopause have ↓ in size of uterus and ovaries and ↓ in vaginal secretion
Skin	The turnover rate of cells in the stratum corneum ↓ with age (e.g., 50% longer to reepithelialize blistered skin at age 65 years compared to the young) ↓ in epidermal cell growth and division and ↓ in number of dermal blood vessels contributes to risk of pressure ulcers Less activity of sweat glands, hydration is difficult to assess by skin alone
Hearing	Presbycusis: high-frequency sounds become harder to hear along with difficulty in distinguishing the changes in tone and speech due to loss of hair cells and fibrous changes in the cochlea Difficulty having a conversation in large, crowded rooms with background noise Conductive hearing loss from middle or outer ear canal (e.g., cerumen accumulation in external canal)
Eyes/vision	Dry eyes from ↓ tear production Three common disease conditions (cataract, glaucoma, macular degeneration—central vision loss)

<27 kg and <16 kg for males and females, respectively;
 (2) muscle quantity: appendicular skeletal muscle mass/height²
 <7.0 kg/m² and <5.5 kg/m² for males and females, respectively;
 and (3) physical performance: gait speed <0.8 m/sec and short
 physical performance battery (SPPB) ≤8 score.

Sarcopenia increases the risk of falls and fractures, impairs mobility and the ability to perform activities of daily living (ADLs), contributes to loss of independence, and increases mortality. Sarcopenia is also associated with cardiorespiratory disease and cognitive decline. The estimated social and economic costs

associated with muscle weakness was £2707 per person per year in the United Kingdom³⁶ and twofold higher for those with sarcopenia in the Czech Republic.⁴⁷ Individuals with sarcopenia had five times higher cost of care during hospitalization compared to those without.⁸

The pathophysiology underlying sarcopenia is still being discovered. Skeletal muscle is the largest reserve of protein in the body and adapts its mass and function to both physiologic and pathologic stimuli.³¹ Mitochondrial metabolic dysfunction, inflammation, and apoptosis were identified as contributors to sarcopenia. Sestrins were recognized as metabolic regulators that affect antioxidant functions and bring about metabolic benefits by nutritional and environmental cues along with cell growth, metabolism, and inflammation. There is increasing evidence that sarcopenia actually originates in early life, including the decline in grip strength across the lifespan.¹⁴

In spite of the serious impact on health outcomes of sarcopenia, recognition in clinical practice is suboptimal. Most clinicians are unaware of the sarcopenia concept or the guidelines for its diagnosis and treatment. The strength, assistance with walking, rising from a chair, climbing stairs, and falls (SARC-F) questionnaire is a validated patient-reported screening tool used to detect signs and symptoms of sarcopenia (Table 31.2).²⁹ The SARC-F questionnaire has five questions asking about difficulty in performing mobility and ADLs with a possible score range of 0 to 10. The recommended cutoff value is 4 points or more for identifying people who are at risk of sarcopenia-associated adverse outcomes.²⁹

The mainstay of interventions for sarcopenia are resistance exercise (RE) and optimizing nutritional intake. RE is shown to improve function significantly. The effectiveness of RE in improving body composition outcomes, however, is less impressive. The prescription of RE is based on specificity, overload, and progression. Monitoring the individual's response using rate of perceived

exertion (0–10 scale) is essential, starting with a score of 3 to 5 and progressing to 6 to 8.²³ Exercise frequency is usually twice a week; however, some individuals may need to start once a week due to a low fitness level. Each session may include two to three sets of 6 to 12 repetitions for lower limb muscle groups or upper limb muscle groups. Intensity of exercise of initial training can be 30% to 60% of 1 repetition maximum (1RM) and advanced to 50% to 70% of 1RM or even higher. The rest interval between training sessions is a minimum of 48 hours. The duration of the exercise program to bring clinical effectiveness is at least 12 weeks. There are barriers identified for older adults to participate in RE programs. The key is to start gently for older adults to experience the benefits of RE without being put off by pain, fatigue, and negative impressions of RE.¹¹

Future directions include the use of routine health data, prescription of REs, translation of biologic interventions in human subject studies, and expansion of the interventions in low- and middle-income countries through global leadership initiatives on sarcopenia. GLIS has been supported by the Asian Working Group for Sarcopenia, EWGSOP, Sarcopenia Definitions and Outcomes Consortium, and 15 major organizations and societies with interest in the field.

Frailty

Frailty is a multidomain and dynamic condition defined as a state of increased vulnerability resulting from age-associated declines in reserve and function across multiple physiologic systems, such that the ability to cope with everyday or acute stressors is compromised.¹⁹ The components of the Fried frailty phenotype operational definition have been widely used. The five components are:

- Unintentional weight loss (≥ 10 pounds in the last year)
- Self-reported exhaustion
- Weakness (grip strength)
- Slow gait speed
- Low level of physical activity

Each component is scored 0 for no and 1 for yes, with the score ranging from 0 to 5. Total score of 0 is categorized as being robust, 1 to 2 as prefrail, and 3 or more as frail.

Osteoporosis and Frailty Fracture

Osteoporosis is a skeletal disease characterized by low bone mass and microarchitectural deterioration of bone tissue with a consequent increase in bone fragility and susceptibility to fracture. Type I osteoporosis is seen in females and is thought to be estrogen-dependent accelerated bone loss after menopause. In this situation, resorption exceeds the rate of bone formation, possibly due to increased cytokine production and subsequent promotion of osteoclastic activities in the estrogen-deficient state. Type II osteoporosis occurs with aging and affects both sexes. Here, the mechanism is due to the decreased number and activity of osteoblasts, which leads to decreased bone formation.

Osteoporosis can be defined as reduced bone strength and bone mass per unit volume that increases risk for fracture secondary to little or no trauma. The degree of osteoporosis is defined by bone mineral density (BMD), which is a value from a test measuring the amount of calcium in a particular region of bone in g/cm³. This value estimates the strength of bone and the amount of bone mass. Among the ways to measure BMD are dual x-ray absorptiometry scan, quantitative computed tomography, quantitative ultrasound, and iliac crest bone biopsy. The World Health

TABLE 31.2 SARC-F Questionnaires and Scoring

Component	Question	Scoring
Strength	How much difficulty do you have in lifting and carrying 10 pounds?	None = 0 Some = 1 A lot or unable = 2
Assistance in walking	How much difficulty do you have walking across a room?	None = 0 Some = 1 A lot, use aids, or unable = 2
Rise from chair	How much difficulty do you have transferring from a chair or bed?	None = 0 Some = 1 A lot or unable without help = 2
Climb stairs	How much difficulty do you have climbing a flight of 10 stairs?	None = 0 Some = 1 A lot or unable = 2
Falls	How many times have you fallen in the past year?	None = 0 1–3 falls = 1 ≥ 4 = 2

SARC-F, Strength, assistance with walking, rising from a chair, climbing stairs, and falls. From Malmstrom TK, Miller DK, Simonsick EM, et al. SARC-F: a symptom score to predict persons with sarcopenia at risk for poor functional outcomes. *J Cachexia Sarcopenia Muscle*. 2016;7(1):28–36.

Organization's definition of osteoporosis is BMD 2.5 standard deviations (SD) below the average value for a healthy young adult.⁵⁶ Osteopenia is categorized as 1.0 to 2.5 SD below the average. A T-score is used for a female in postmenopause or a male aged 50 years or older and is the difference between the patient's BMD and zero, which is the BMD of a healthy young adult.⁵¹ A Z-score is used for premenopausal females or males younger than 50 years. It is the difference between the patient's BMD and the average BMD for healthy people of similar age, ethnicity, and sex. Osteoporosis is one of the most common bone diseases in humans. Worldwide, 9 million fractures due to osteoporosis occur annually. One in three females and one in five males over age 50 years will experience osteoporotic fractures.⁵⁶ Approximately one-third of patients with a hip fracture will fracture the opposite hip.

Indications for BMD screening from the US Preventive Services Task Force are:

- Females age 65 years and older should be screened routinely for osteoporosis (grade B recommendation).
- Postmenopausal females younger than 65 years at increased risk of osteoporosis should be screened for osteoporosis.⁵²

Osteoporotic or frailty fractures have varying definitions but can be thought of as fractures that occur in individuals with low BMD.⁵⁶ Common sites of osteoporotic fractures are the hip, spine, pelvis, sacrum, and humerus. Risk factors for osteoporotic fractures include falls, advanced age, dementia, reduced physical activity, female sex, and White race. Factors contributing to increased risk for osteoporotic fractures include low BMD, previous osteoporotic fracture, increasing age, maternal history of fracture, certain medications, small body habitus, sedentary lifestyle, menopause, and estrogen levels.⁵⁶

Osteoporosis is classified as either primary or secondary osteoporosis. The classification is based on factors affecting the metabolism of bone. Primary osteoporosis is osteoporosis that develops as a result of aging or menopause-related bone demineralization. Type I/postmenopausal and type II/senile osteoporosis are two subtypes of primary osteoporosis. Secondary osteoporosis is the result of medications or pathologic conditions other than aging and menopause that lead to decreased bone mass and increased fracture risk.³

Laboratory tests are often ordered to exclude secondary causes of osteoporosis in patients. Recommended laboratory tests for evaluating osteoporosis include complete blood count, thyroid-stimulating hormone, vitamin D, creatinine, hepatic transaminase, calcium, phosphorus, alkaline phosphatase, albumin, erythrocyte sedimentation rate, and 24-hour calcium excretion level.⁴⁶

Nonpharmacologic interventions to address osteoporosis target diet, exercise, smoking cessation, reduction of alcohol intake, and maintenance of an ideal body weight.⁵⁶ Common pharmacologic treatments are bisphosphonates, selective estrogen receptor modulators, calcitonin, parathyroid hormone, estrogen replacement therapy, hormone therapy, denosumab, fluoride, calcium, and vitamin D. The FRAX algorithm predicts the 10-year fracture probability using numerous factors and information. It considers the patient's country, BMD, age, sex, and clinical risk factors (e.g., personal history of fracture, low body mass index, use of oral glucocorticoid therapy, rheumatoid arthritis, osteoporosis secondary to another condition, parental history of hip fracture, current smoking, and alcohol intake of three or more drinks per day).^{18,35}

Falls Prevention and Intervention

Falls are defined as a loss of upright position that results in downward displacement of the body to the floor or hitting another

object.³⁵ Falls are a significant issue for older adults. It is estimated that one of three older adults age 65 years or older falls each year. By 80 years of age, over half of older adults fall annually.⁵⁵ Falls have direct costs, such as health care costs, and indirect costs, such as societal productivity. Hospital inpatient costs account for over half of the cost of falls, followed by long-term care costs. Indirectly, falls can cause lost productivity of the individual and family/caregivers, leading to lost income. Falls are often multifactorial (Fig. 31.1) but can be broken down into the following categories: environmental, physical, mental, and pharmacologic. Table 31.3 shows examples of interventions for the different categories.

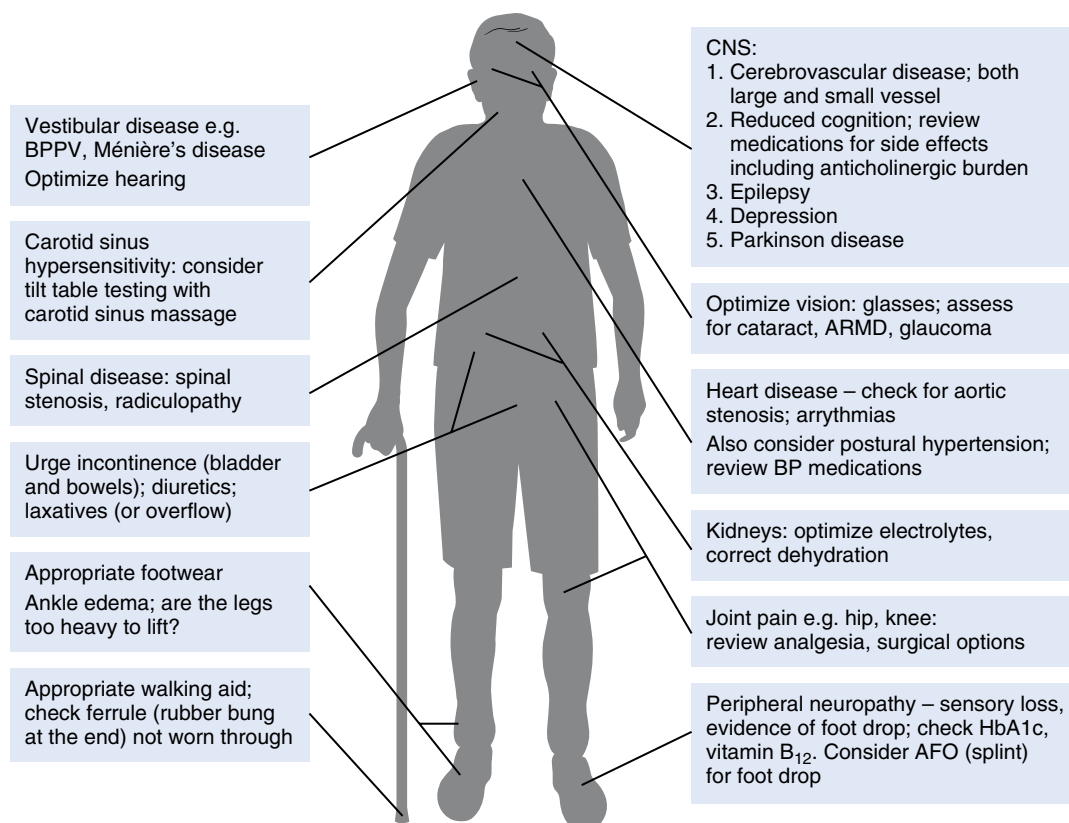
Arthritis and Joint Replacement

Osteoarthritis is the most common musculoskeletal condition that causes joint pain, and as the population ages, it becomes more prevalent. Of adults age 65 to 74 years, 44% have arthritis, and 53.9% of adults aged 75 years and older are found to have arthritis.¹⁵ Treatment is initially conservative, but many geriatric patients will eventually receive a joint replacement. Conservative management consists of nonsurgical treatment options such as patient education, weight loss, therapeutic exercise, assistive devices, and orthoses. It also consists of pharmacologic interventions, such as oral pain medications, topical pain medications, joint injections, nerve blocks, and ablations.⁴

Patients who have failed conservative management and are good surgical candidates will often proceed to joint replacement. Surgical indications include failure of conservative treatment with severely limiting joint pain, decreased mobility, and decreased quality of life. Surgical contraindications include infection, medical risk factors, osteoporosis, dementia, young age, and limb weakness. Elderly patients should be counseled appropriately on the risks associated with surgical intervention. Compared to younger patients, older total hip arthroplasty and total knee arthroplasty recipients were more likely to experience postoperative complications, admission to the intensive care unit, discharge to a skilled care facility, and longer hospital length of stay.¹⁶ Patients should undergo a thorough preoperative assessment, including a close assessment of their cardiopulmonary status. Perioperatively, optimal control of medical comorbidities is needed. Postoperatively, early mobilization, therapy, and pain control can significantly impact outcomes.¹⁶ Postoperative rehabilitation is key to recovery after surgery. It can occur in various settings and intensities. Options include acute inpatient rehabilitation, subacute rehabilitation, home therapy, and outpatient therapy. It has been shown that similar results can be achieved in all settings with proper resources and coordination of care.²⁸

Neurologic Disorders Affecting Gait and Balance

Human walking is a complex interplay of nervous, musculoskeletal, and cardiorespiratory systems and requires cognitive executive function for planning along with continuous interaction with the environment.^{37,44} Testing walking during dual tasking shows the taxing effect of cognition on gait in older adults and individuals with central nervous system disorders. Gait speed is often measured as a quantitative metric of abnormal gait. Gait speed declines by 1% per year from 60 to 80 years of age.⁹ Slow gait speed has been shown to predict functional decline, dementia, and mortality in older adults. The prevalence of abnormal gait is



• **Fig. 31.1** Systems approach to assessing falls risk factors. AFO, Ankle-foot orthosis; ARMD, age-related macular degeneration; BP, blood pressure; BPPV, benign paroxysmal positional vertigo; CNS, central nervous system. (From Harper A, Wilkinson I. Falls in older adults: causes, assessment and management. *Medicine*. 2021;49[1]:32-37.)

TABLE 31.3 Fall Risk Assessment and Interventions

Category	Intervention
Environmental	Home assessment, floor mats, hip protectors, self-locking wheelchairs
Mental	Cognition screens such as Mini-Mental State Examination, Geriatric Depression Scale, and Montreal Cognitive Assessment
Physical	Hearing and vision screens, vitamin levels, footwear, compression stockings; gait and balance assessment, Tinetti fall eval, Morse fall risk assessment, stratify fall risk assessment, exercise, range of motion/flexibility, balance exercises
Pharmacologic	Screening, side effects, Beers criteria, reduce key classes where possible, antihypertensives, diuretics, pain medications, antiepileptics, antidepressants, antipsychotics, steroids, medication reviews

32.2% among adults aged 60 to 97 years, one-quarter of which were neurologic gait abnormalities.²⁷ However, 38.3% of those aged 80 years or older still had normal gait. Therefore gait disorders are not an inevitable part of aging. The etiologies underlying the development of abnormal gait include neurologic, musculoskeletal, medical (e.g., cardiopulmonary, obesity), and medication-related

issues. Treatment approaches aim to address the underlying disease with medical and operative interventions combined with gait training, assistive devices, and education regarding falls.

There are a number of specific abnormal gait patterns among older adults. Hypokinetic rigid gait is characterized by shuffling with short step length, a stooped posture, and turning en bloc. Management includes optimizing medications targeting specific conditions if present (e.g., Parkinson disease), addressing orthostatic hypotension, and prescribing rehabilitation programs. Spastic gait can be hemiparetic or paraparetic. Spastic hemiparetic gait is caused by unilateral brain dysfunction with typical extensor synergy pattern, circumduction of the swinging leg, and slow gait speed. Spinal cord lesions, including cervical myelopathy from spondylosis, may cause spastic paraparetic gait. Ataxic gait can be from cerebellar lesions or proprioceptive sensory disturbances. Cautious gait refers to an excessive degree of age-related change in walking and fear of falling. Walking difficulties are often out of proportion to the level of sensory and motor dysfunction. Gait often improves with very minimal support, including touching of the patient's hand.³⁷ Medication use can cause gait impairment, including antidepressants (selective serotonin reuptake inhibitors and tricyclic antidepressants), benzodiazepines, analgesics, anticholinergics, and anticonvulsants.⁴⁴ Medication reconciliation should be a mandatory part of evaluation of gait disorders in older adults. Table 31.4 summarizes the gait characteristics and underlying causes of common gait disorders in older adults.

TABLE 31.4 Abnormal Gait Patterns Among Older Adults

Abnormal Gait	Gait Characteristics, Associated Features	Underlying Causes
Hypokinetic rigid gait	Slow speed, short step length, shuffling, freezing	Parkinson disease Dementia with Lewy bodies Frontal lobe lesions (e.g., mass)
Ataxic gait	Staggering, wide based	Cerebellar lesions, sensory proprioceptive disturbances
"Magnetic gait"	Wide based, freezing, preserved arm swing, foot appeared stuck on the ground by "magnetic force" Urinary incontinence, cognitive decline	Normal pressure hydrocephalus
Spastic hemiparetic gait	Wide or narrow based depending on the severity of spasticity, slow gait speed, extensor synergy with knee extended and foot plantarflexed inverted	Unilateral hemispheric lesions of the brain
Spastic paraparetic gait	Narrow based, adducted legs, slow gait speed Hand incoordination	Spinal cord lesions (e.g., cervical myelopathy)
Cautious gait	Disproportionate degree of gait disturbance relative to preserved sensory, motor, and cognitive function. Gait often improves with minimal support (e.g., touching of hand)	Anxiety, fear of falling

Age-Friendly Care in Rehabilitation

Age-friendly care has become an area of focus throughout the medical field as the proportion of Americans over the age of 65 years continues to grow. The 2020 US census data demonstrated that 16.8% of the population was over age 65 years, representing about 55.8 million people. This population is expected to increase to about 95 million individuals by 2060.⁴⁹ Health systems have sought to determine a guideline approach to caring for older adults throughout the continuum of medical care, and psychiatry plays a crucial role in this care. The Institute for Healthcare Improvement (IHI) has developed the Age-Friendly Health Systems initiative, which seeks to recognize institutions that provide evidence-based care to the geriatric population within a framework of the 4Ms: Medication, Mentation, Mobility, and What Matters to You.²⁴ This framework can be a complementary guide for the psychiatric care of older adults in both inpatient and outpatient settings.

Medication Usage

Metabolism

With increasing age comes changes in the pharmacokinetics of medications. Medication absorption can be impaired by the age-related decrease in small bowel surface area, slowing of gastric emptying, and increase in gastric pH. This decreased acidity can cause the early release of enteric-coated medications. The relatively increased fat volume of older adults also increases the distribution of lipophilic medications, increasing their half-lives and potential for toxicity. Hepatically metabolized medications through the cytochrome P450 system also require titration in the geriatric population, as first-pass metabolism has been shown to decrease by about 1% per year after age 40 years. Renal excretion diminishes with age as the glomerular filtration rate also declines, causing the need for close monitoring of renally metabolized pharmaceuticals.⁴⁰

Medication Reconciliation

Each encounter with patients is an opportunity to review their medication profile to deprescribe, downtitrate, or further educate

them about proper use of their medications. In the inpatient setting, this can be done in conjunction with the rehabilitation team's pharmacist. Polypharmacy is typically defined as taking five or more prescription drug treatments and is associated with increased mortality, disability, hospitalizations, falls, cognitive and functional impairment, adverse drug events, and worsened adherence to a medication regimen.^{22,39} As of 2018, about 44% of the US geriatric population fall into this category, which has been trending upward.⁵⁴ Particular attention should be paid to medications that have central nervous system or cognitive effects, such as antidepressants, opioids, antiepileptics, benzodiazepines, medications with anticholinergic properties, and sleeping pills. These should be evaluated in the context of necessity, renal and hepatic function, efficacy, and possible interactions with the patient's other medications.

The Beers criteria is a set of guidelines created by the American Geriatrics Society that is regularly updated and can help guide medication choices in the geriatric population. The guideline specifies potentially harmful medications for older adults, medications with significant side effects associated with certain health conditions, and medications that interact with other medications commonly taken by this age group. This guideline can be utilized during a reconciliation to advise prescribing choices.⁶

Pain Management

Pain, both acute and chronic, is addressed by psychiatrists in both the inpatient and outpatient settings. In the geriatric population, pain is associated with reduced mobility, increased anxiety and depression, and disruption of social relationships.²⁰ Pain management should be approached in a multidisciplinary fashion with escalation from nonpharmacologic (physical and occupational therapies, early mobilization, environmental modifications, psychological interventions) to pharmacologic and, finally, invasive interventions. When pharmacologic therapy is considered, noncentrally acting medications should be utilized when indicated. For example, providers should consider the use of calcitonin for vertebral compression fractures¹⁰ rather than opioids or nonsteroidal antiinflammatory drugs (NSAIDs). When opioids or

NSAIDs are the indicated treatment of choice, consideration should be given to the difference in pharmacodynamics and kinetics noted earlier, which may lead to decreased tolerance and an increased risk for adverse drug reactions. For opioids, this includes constipation, sedation, and increased falls risk. NSAID use in older adults is associated with gastric bleeding and ulceration, renal dysfunction, myocardial infarction, and stroke.²¹

Mentation

Delirium

Delirium is commonly encountered in the inpatient acute care and rehabilitation settings. It is defined as “an acute, fluctuating syndrome of altered attention, awareness, and cognition” often seen in the geriatric population.²⁵ Delirium has up to a 60% prevalence in the acute rehabilitation setting, with many cases likely going undiagnosed.²⁵ Delirium can prevent meaningful participation in a rehabilitation program and cause significant decline in function. Addressing delirium starts with prevention techniques that all members of the care team can help to employ (Box 31.1).

Diagnosis of delirium can be made using criteria from the *Diagnostic and Statistical Manual of Mental Disorders*, Fifth Edition.⁷ Screening for delirium should occur on initial patient encounter and periodically throughout the patient’s rehabilitation stay. The Brief Interview for Mental Status is a tool used to determine a baseline level of cognition by quickly assessing a patient’s repetition, temporal orientation, and recall. It can be repeated to show any fluctuation in this baseline that may raise concern for developing delirium or, on a longer term, existence of dementia.³⁰ The Confusion Assessment Method is a validated instrument and diagnostic algorithm for the identification of delirium and is available in both long (10 questions) and short (4 questions) forms. Together, these tools can be used by any member of the care team to potentially identify delirium early. Workup and treatment for delirium should include medication reconciliation and assessment of causative etiologies, such as infection or electrolyte derangements. Treatment of the underlying cause is the initial goal followed by supportive care and prevention of worsening until delirium clears. This involves ensuring adequate nutrition and hydration, enforcement of sleep patterns, assuring pain management, and, if absolutely necessary, pharmacologic management with antipsychotics, benzodiazepines, or antidepressants.²⁵

• BOX 31.1 Delirium Prevention Recommendations

- Promote periods of continuous sleep
- Use frequent reorientation techniques
- Promote moving patient out of bed at least three times a day for meals when possible
- Encourage visual and hearing aids if there is any impairment present
- Encourage family members at bedside
- Avoid medications that may worsen delirium
 - Anticholinergics
 - Antihistamines
 - Benzodiazepines
 - Opioids
- Avoid physical restraints unless patient is a danger to self or others
 - Utilize one-to-one observation if necessary
- Remove indwelling catheters and minimize lines as much as medically possible

Dementia

Dementia is a collective term describing a pathology that affects memory, cognition, and communication and causing increased difficulty with motor and coordination functions. The most common types of dementia are Alzheimer disease, vascular dementia, Lewy body disease, frontotemporal lobar degeneration, and Parkinson dementia.² These are typically considered irreversible and often progressive. The role of the psychiatrist in dementia involves early recognition of its onset and the development of an intervention plan to address the progressive deficits the patient will face. This often involves a multidisciplinary team consisting of physical and occupational therapists, speech therapists, and neuropsychologists. Together, cognition, language, and mobility can be bolstered by the team approach prior to the patient’s decline, or compensatory strategies can be developed once skills and abilities begin to diminish.

Depression

Depression can develop acutely in the geriatric age group or continue chronically from early ages. Depression leads to worsened in-hospital health status and lower rates of improvement both in the acute and postacute settings.¹² At times, the causes of this depression are related directly to musculoskeletal impairments leading to lack of mobility, pain, and loss of independence. The psychiatrist can play a direct role in treating all these problems. It is also essential to ensure patients are provided with appropriate referrals for follow-up with a mental health professional.

Mobility

Maintaining mobility is essential for healthy aging because it has direct beneficial effects on physical, neurologic, and mental health quality of life in the geriatric population. Physiologic changes in the elderly, frailty, and falls are key components of geriatric mobility and are discussed elsewhere in this chapter.

Mobility Screening

Mobility can be assessed using a number of validated screening measures. Examples include the timed up and go test, SPPB, dynamic gait index, 6-minute walk test, 5-time sit to stand test, and Berg balance scale.⁴⁵ These can be performed in both inpatient and outpatient settings by psychiatrists, nursing, or therapy teams. The clinician’s choice of test depends on the purpose of the assessment, such as determining risk of falls at home, assessing the ability to perform a task-specific activity, or measuring endurance. These measures exist in the psychiatrist’s toolbox for patient assessments, and outcomes can be tracked over time to guide treatments and recommendations.

Adaptive Equipment

Durable medical equipment is used to address mobility concerns in all age groups for both chronic and acute causes of mobility limitations. In the elderly, equipment recommendations should consider the patient’s individualized needs and abilities, including coordination and cognitive function, home environment, available assistance, and insurance coverage. For example, if a walker is deemed necessary within the home, consideration should be given to the patient’s ability to negotiate stairs in the home while toting the walker. If unable, the patient should be recommended to have a walker on each floor of the home. Straight single-point canes or quad canes are relatively lightweight and convenient devices that are used in one hand and provide proprioceptive feedback and

increased stability while offloading a potentially painful hip or knee. Walkers provide a greater degree of stability but require the use of both arms and have increased weight and size when compared to a cane. A rollator can be useful in the more mobile patient because it provides a consistent place to sit and rest when needed. However, rollators can cause falls if the brakes are used incorrectly or if the device, which is on wheels, moves too quickly for the patient to keep up with. A wheelchair should be considered to allow for longer distance mobilization outside the home. Reliance on a wheelchair can lead to disuse atrophy and further immobilization; thus active mobility should always be encouraged. A combination of adaptive devices is often the correct solution for short- vs. long-distance ambulation as well as home vs. community ambulation.

Orthotics

The use of foot orthoses in the elderly is helpful for the treatment of foot, leg, or back problems. They redistribute pressure, provide cushion, and modify alignment of the feet and legs. They have been shown to improve postural stability and lower extremity function. This provides an inexpensive and easy-to-implement recommendation that improves balance and reduces falls. Foot orthoses can be acquired over the counter or through custom molding, which would require referral to a podiatrist.³³

What Really Matters

Arguably, the most important M in the 4Ms framework is determining what matters to each individual patient. Psychiatrists, through their innately holistic approach to patient care, are well positioned to guide patients in identifying their needs, preferences, and goals. Health-related quality of life can be assessed using several different instruments and is fundamentally patients' perception and evaluation of their own health and its effect on their daily living. It can be useful to identify areas of potential improvement that psychiatrists can target using the interdisciplinary team at their disposal. This may include mobility concerns, for which physical and occupational therapy programs can be crafted, or pain, for which pharmacologic or noninterventional modalities can be prescribed. Through these assessments, psychiatrists can help enhance the overall quality of life for geriatric patients.

Exercise Recommendation and Sports Injuries in Older Adults

The Centers for Disease Control and Prevention recommends that every week, adults age 65 or older need to engage in (1) aerobic physical activities of at least 150 minutes of moderate intensity (e.g., 30 min/day, 5 days/week) or 75 minutes of vigorous intensity or equivalent combination of moderate intensity and vigorous intensity, (2) at least 2 days of activities that strengthen muscles, and (3) activities to improve balance.⁵⁰ Multicomponent physical activities with all three components are some sports, tai chi, and gardening. As the population ages, the number of aging athletes also increases. Older adults report that the main drivers for playing sports are maintaining their health and well-being, socializing and feelings of positivity, developing new relationships and strengthening existing ones, competition, and achievement.⁴⁸ Pickleball is the fastest-growing sport in the United States, according to the Sports and Fitness Industry Association, and it is joining the list of age-friendly sports, such as

golf and swimming. The strategies to prevent sports-related injuries include maintenance of joint flexibility, allowing sufficient recovery intervals between plays, and paying attention to the early signs of injuries ("listen to your body"). Unusual discomfort, tingling, numbness, weakness, or limited range of motion are early signs of potential injuries or disorders. In addition, older adults who remain flexible and open-minded about the choice of sports and are willing to consider transitioning to relatively low-impact sports seek to continue enjoying sports activities for their lifetime. The American Council on Exercise recommends drinking 17 to 20 ounces of water 2 hours before working out, 7 to 10 ounces every 10 to 20 minutes, and 16 to 24 ounces for every pound of body weight lost.⁵

Assistive Technologies for the Geriatric Population

Assistive technology (AT) is defined as any equipment, software program, or product system that is used to improve the functional capabilities of people with or without disabilities. AT includes both low- and high-tech solutions. Often older adults and their caregivers do not have an adequate knowledge of what ATs are available, how to use them, and where to find training and support.³⁸ The Assistive Technology Act Technical Assistance and Training Center (AT3 Center) was created as a sponsored project of Assistive Technology Act Programs with the mission of increasing access to and acquisition of AT by individuals with disabilities across their lifespan.³⁴ The AT3 Center website (<https://at3center.net/>) lists the resources about the AT Act Program in each state. Low-tech AT includes tools offered to improve functional mobility or ADLs: chair lift with handles, walking aids, reacher, toilet aids, and bed handles. High-tech devices include alarm monitoring devices for medical emergencies, telehealth systems with sensors monitoring vital health information and communicating with providers, GPS trackers for individuals at risk for wandering, and various memory aids reminding patients about essential items (e.g., keys), taking medications, or feeding their pets.

There are many personal alarm products for people to call for help. In choosing the right device for an individual, one should consider the feature that allows the wearer to talk to the monitoring team through the device, automatic fall detection, GPS capability, the requirement for internet or landline connectivity, battery life, and the cost structure (rental, monthly fee).

Smart home technology allows users to program and control household appliances, door locks, and other devices at home. Voice control adds convenience to smart home systems. Several states have smart home demonstration sites, including North Dakota, Oklahoma, Illinois, and Alaska. Smart home monitoring systems with home sensors can detect any unusual activities or inactivity of people in their homes and update the users' caregivers. Additional features include the locking of gates, virtual monitoring of users, and specific notifications for opening medication cabinets. Several ATs for cognitive dysfunction have been studied with mixed results. For example, a pilot trial of a wearable camera that informs the user of a visitor's name and relationship found no evidence of improvements in social engagement or quality of life.³² Social stigma, AT reliability, and system feasibility have been identified as barriers to meaningful use.³²

The American Association of Retired Persons (AARP) featured several technologies that may help people maintain their independence in their own homes for the future: (1) artificial intelligence

to answer questions, (2) telehealth, (3) smart cars with driver safety technologies and blind-spot warnings, and (4) robots to assist around the house (e.g., Labrador Retriever). The AgeTech Collaborative from AARP was launched with a global mission to improve people's lives as they age.¹ Through this startup collaboration, forward-thinking investors, enterprises, and testbeds come together to bring user-centered technology innovation to the market.

Ethical Decision Making

Biomedical ethics is the branch of applied ethics that studies the philosophical, social, and legal issues arising in medicine and the life sciences. It is comprised of four principles⁵³:

- **Autonomy:** Patients have the power to make rational decisions and moral choices, and each patient should be allowed to have self-determination.
- **Nonmaleficence:** The physician is obligated to not harm the patient.
- **Beneficence:** The physician is obligated to act for the patient's benefit.
- **Justice:** Patients should receive fair, equitable, and appropriate treatment.

Biomedical ethics issues often arise when these principles come into conflict. Factors such as informed consent and capacity can also play a part in ethical conflicts. Many hospitals have ethics committees and ethics consults available to support resolution of these conflicts. In geriatric patients, these conflicts often come to the forefront during end-of-life care.

The ethical decision-making model at a glance consists of¹⁷:

1. Identify the problem.
2. Apply the American Counseling Association's Code of Ethics.
3. Determine the nature and dimensions of the dilemma.
4. Generate potential courses of action.
5. Consider the potential consequences of all options and determine a course of action.
6. Evaluate the selected course of action.
7. Implement the course of action

Summary

Care of the geriatric population has never been more important, and it is critical that the psychiatrist understand some of the specific problems that commonly impact this group. Sarcopenia, frailty, osteoporosis, falls, arthritis, and neurologic conditions are

important causes of functional difficulties, and the Age-Friendly Health System Initiative's 4Ms (Medication, Mentation, Mobility, and What Matters to You) can be a very helpful framework to use in clinical practice. Using these principles and others outlined in this chapter, physical medicine and rehabilitation providers can help maintain and improve the quality of life for older adults.

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